Table 5: Compare qwen2.5-7B, 32B, with the results of reclassification of the model we trained on qwen2.5-7B for the sentiment, intent, theme, and style of user input

Model	RE-classification Accuracy↑			
	Emotion	Intention	Theme	Style
Qwen2.5-7B-Instruct	0.420	0.515	0.551	0.521
Qwen2.5-32B-Instruct	0.571	0.611	**0.616**	0.603
HUMOR-CoT	**0.597**	**0.641**	0.600	**0.639**

improves from 0.1944 (threshold 0.5) to 0.3888 (threshold 0.9), alongside a corresponding increase in accuracy.

Human judgments vs. true labels. Human judgments show weak negative agreement with ground-truth authenticity (Cohen's $\kappa = -0.4397$, $p < 0.05$; Spearman $\rho = -0.4493$, $p < 0.05$), likely due to anthropomorphism and the difficulty of discerning AI-generated memes.

Conclusion. Gemini aligns with ground-truth labels substantially better than human annotators, and its continuous outputs encode meaningful gradients of human-likeness. These findings, combined with its high specificity and top AUC, support using Gemini-2.5-pro as a reliable evaluator for Human Rate.

K.5 Human Alignment of Gemini's Group-wise Ranking Evaluator

To validate the relative ranking results in Fig. 4(b), we perform an independent human evaluation that mirrors the same group-wise comparison protocol described in Sec. 4.2. For five representative image groups (30 memes total), nine human annotators ranked the six meme variants—*HUMOR-CoT*, three CoT baselines, *In-the-wild*, and *Text-Free*—under the same criteria rubric used by *Gemini-2.5-pro*. Each annotator produced one holistic ranking per group.

We compute rank correlation between Gemini's and human rankings. Across the five groups, Gemini exhibits strong and consistent alignment with human preferences, with a mean **Spearman correlation of 0.7188 $\pm$ 0.2154** and **Kendall's τ of 0.6320 $\pm$ 0.2269**.

These results confirm that the group-wise ranking in Fig. 4(b) captures preference structures also expressed by humans, providing quantitative evidence that the relative VLM evaluation is meaningful and not an artifact of evaluator noise. Combined with the preceding analyses, this supports the reliability of the Gemini-based ranking methodology used throughout our evaluation.

L Supplementary Experiment Results

L.1 VLM Classification Result

To further examine whether our generated meme texts faithfully reflect the intended semantics of user input, we perform a reclassification experiment using a strong vision-language model (VLM) as an external evaluator. Specifically, we take the captions generated by each model and feed them into the same VLM classifier that was trained to recognize four major semantic axes: *emotion*, *intention*, *theme*, and *style*. The classifier outputs predicted labels for each axis, which are compared to the original user-specified categories to compute reclassification accuracy.

Table 5 summarizes the results. HUMOR-CoT achieves the highest accuracy across all dimensions, surpassing both the Qwen2.5-7B-Instruct and the larger Qwen2.5-32B-Instruct baselines. This indicates that our hierarchical CoT fine-tuning not only improves humor expressivity but also enhances the faithfulness of generated texts to user intent. In particular, the improvement over the 32B model suggests that structured reasoning contributes more effectively to semantic alignment than mere parameter scaling.

L.2 Cross-Domain Pilot on Persona-Conditioned Visual Captioning

To examine whether the HUMOR-style reasoning and group-wise selection procedure can be instantiated beyond meme generation, we conducted a small cross-domain pilot under the rebuttal timeline rather than a full transfer benchmark. We use persona-conditioned visual captioning based on Personality-Captions [Shuster et al., 2019]. Each group is defined by the same recovered image and the same personality trait, i.e., $G = (\text{image}, \text{personality})$. This setting matches the formulation in Sec. 2: outputs are open-ended captions, quality depends on subjective criteria such as engagement, persona consistency, visual grounding, and coherence, and pairwise comparisons are naturally meaningful within the same image-persona group.

We sampled 20 recovered image-persona groups and used the same Qwen2.5-VL-7B-Instruct model for both methods. The direct baseline generates four captions directly. The HUMOR-style procedure first explores visual evidence, persona intent, and multiple plausible caption angles, and then generates four candidates under the same image-persona condition. Figure 15 provides representative qualitative examples.

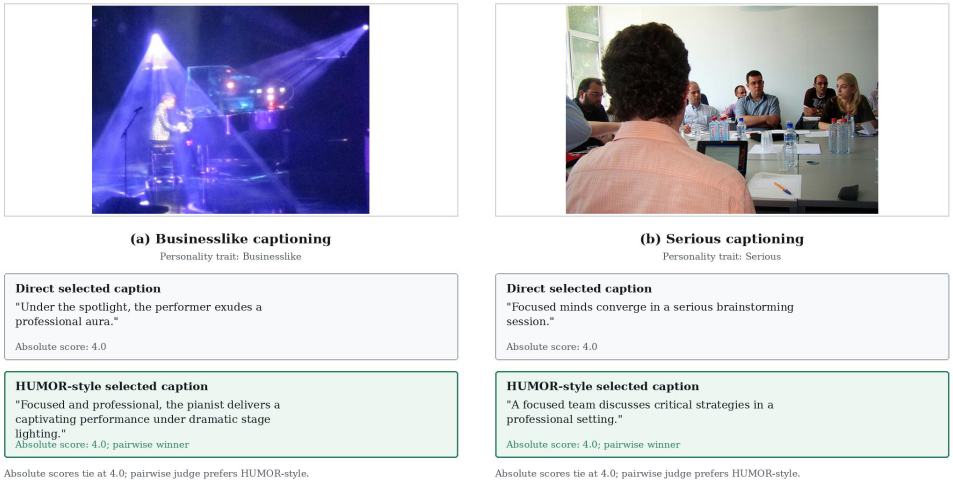


Figure 15: Qualitative examples from the persona-conditioned visual captioning pilot. For the same image-persona group, HUMOR-style multi-path reasoning explores different caption strategies before local selection.

Table 6: Candidate diversity in the cross-domain pilot. distinct-1 measures word-level diversity, distinct-2 measures local bigram-level diversity, and average pairwise sentence distance measures semantic diversity in sentence-embedding space.

Method	#Groups	distinct-1	distinct-2	Avg. sentence dist.
Direct	20	0.734	0.960	0.507
HUMOR-style multi-path	20	0.738	0.942	0.563

As shown in Table 6, HUMOR-style generation slightly improves word-level diversity (0.734 $\rightarrow$ 0.738) and more clearly improves semantic diversity (0.507 $\rightarrow$ 0.563, about $+11.0\%$). distinct-2 slightly decreases (0.960 $\rightarrow$ 0.942), so we do not claim uniform lexical diversity improvement. Instead, the result indicates that multi-path reasoning primarily broadens the space of high-level caption strategies: candidates move farther apart in sentence meaning, persona interpretation, and communicative angle under the same image-persona condition, rather than simply swapping local bigrams or increasing surface phrase variation.

We also compare independent absolute rubric scoring with group-local pairwise judging. In absolute scoring, each candidate is scored independently on engagement, persona consistency, visual grounding, coherence, and overall quality. This scoring is not discriminative in our pilot: all 160 candidate

captions receive 4.0 on every dimension. In contrast, group-local pairwise judging provides more informative preference signals.

Table 7: Group-local pairwise preference results comparing the HUMOR-style selected caption against the direct selected caption.

Judger	#Groups	HUMOR wins	Ties	HUMOR losses
Qwen2.5-VL-7B-Instruct	20	14	0	6
MiniMax-M3	20	12	3	5
Step-3.7-Flash	20	10	9	1

Across three judgers, HUMOR-style selected captions are preferred more often than direct selected captions: Qwen gives 14 wins vs. 6 losses, MiniMax-M3 gives 12 wins vs. 5 losses with 3 ties, and Step-3.7 gives 10 wins vs. 1 loss with 9 ties. The Step-3.7 result is weaker because many cases are tied, and this pilot uses only the reasoning and selection component rather than full HUMOR reward-model training. We therefore treat the result as directional evidence rather than a full cross-domain generalization claim. Still, the pilot supports two core points of the framework: absolute scoring can be poorly discriminative for subjective multimodal generation, while local pairwise comparison within semantically coherent groups can provide useful preference signals beyond the meme domain.

M Limitations and Societal Impact

This section provides a self-critical analysis of the HUMOR framework, its underlying model HUMOR-RM, and the potential broader impacts of our work.

M.1 Inference Latency and Computational Cost

As detailed in Section 3.1 and Appendix I, the hierarchical multi-path CoT mechanism requires the model to generate structured reasoning trajectories before yielding the final punchline. While this "thinking" process is essential for the logical leap required by humor, it incurs a computational overhead of approximately $2\times$ to $3\times$ more tokens than standard VLM captioning. This latency may hinder real-time deployment on mobile devices. Future work could explore Early Exit strategies to bypass reasoning for simpler visual stimuli.

M.2 Cultural and Linguistic Specificity

Humor is intrinsically tied to cultural context, slang, and shared history. Our training data [Xu et al., 2024] is primarily composed of English-language internet memes. Accordingly, our empirical results should be read as evidence within an English-speaking meme setting, not as a universal measurement of humor across all cultures. At the same time, the HUMOR framework itself does not rely on language- or culture-specific assumptions: it requires coherent groups and reliable within-group preferences. Extending the data collection, reward validation, and human evaluation to multilingual and multicultural meme communities is an important direction for future work.

M.3 Failure Modes

Despite the anchoring mechanism described in Proposition 1, qualitative reviews (see Appendix N.5 for failure cases) indicate that the model occasionally fails to generate a humorous incongruity when the input image is highly complex or visually cluttered. In these cases, the CoT tends to default to literal description rather than finding a creative twist. This suggests that the model's "creativity" is still bounded by the diversity of its pre-trained visual-linguistic associations.

M.4 Broader Societal Impact and Safety

The ability to automate humor generation carries risks. Memes are frequently used for political satire, which can border on misinformation or harassment. Although we utilized Gemini-2.5-pro as a risk judge (Section 4.3) to filter out hate speech and sensitive content, adversarial prompts could